UNIVERSITY EXAMINATIONS

UNIVERSITEITSEKSAMENS

UNISA university of south africa

MAT1503
MAT103N

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LINEAR ALGEBRA (MATHEMATICS)

Duration 2 Hours

100 Marks

EXAMINERS :
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THE USE OF A POCKET CALCULATOR IS NOT PERMITTED.

This paper consists of 4 pages.

Answer all the questions. There is a total of 100 marks. 100 marks will count as full marks.

QUESTION 1

Consider the following system of linear equations

$$\begin{aligned} x_1 + x_2 - x_3 &= 2 \\ 3x_1 + 2x_2 - x_3 &= 3 \\ -x_1 - x_2 + 2x_3 &= -1 \end{aligned}$$

Write down the **augmented matrix** of the system, reduce the augmented matrix to generalized row-echelon form, and then determine the solution of the system.

[8]

QUESTION 2

Consider the same system of linear equations as in Question 1·

$$\begin{aligned} x_1 + x_2 - x_3 &= 2 \\ 3x_1 + 2x_2 - x_3 &= 3 \\ -x_1 - x_2 + 2x_3 &= -1 \end{aligned}$$

[TURN OVER]

Suppose we write the system in matrix form, that is,

$$A\underline{x} = \underline{b}, \quad \text{where } \underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

2 1 Write down the matrices A and $\underline{b}$ (3)

2 2 What are the respective sizes of matrices A, $\underline{x}$ and $\underline{b}$? (3)

2 3 Determine the inverse A^{-1} of matrix A by applying the **matrix inverse algorithm**, that is, by applying a sequence of elementary row operations on $[A \mid I]$. (8)

[14]

QUESTION 3

Use the results in 2.1, 2.2 and 2.3 and the matrix form $A\underline{x} = \underline{b}$ to determine the solution of the system of linear equations given in both Questions 1 and 2 Include the steps of your reasoning
[**Note** If the solution that you obtain in Question 3 does not correspond to the solution that you determined in Question 1, you must have made a writing or numerical error somewhere. Use the opportunity to find and correct it!]

[6]

QUESTION 4

Suppose we have the square matrix

$$C = \begin{bmatrix} 1 & 1 & -1 \\ 3 & 0 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

4 1 Evaluate $\det(C)$ by expanding along the **first row** Include the steps of the calculation **No marks** will be awarded if you use **any other method** (7)

4.2 Does the homogeneous system

$$C\underline{x} = \underline{0} \quad \text{where} \quad \underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad \underline{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

have no solution, only one solution or infinitely many solutions? Give a reason for your answer. (2)

4 3 Use the value $\det(C)$ that you determined in 4.1 to evaluate the following Include your reasoning.

(a) $\det\left(CC^{-1}\right)$ (2)

(b) $\det(3C)$ (2)

(c) $\det\left(CC^{T}\right)$ (3)

(d) $\det\left(2C^{-1}\right)$ (2)

[18]

QUESTION 5

5.1 You are given six 3×3 matrices

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad F = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad G = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$$

(a) Identify those that are elementary matrices, and then write down only the elementary matrices

(b) Give the general definition of an elementary matrix. (5)

5 2 (a) Is the inverse of an elementary matrix also an elementary matrix?

(b) Determine the inverse matrices of the elementary matrices that you wrote down in 5 1(a) (4)

[9]

QUESTION 6

6.1 Let L_1 be the line passing through the points $A\,(3,0,2)$ and $B\,(4,3,0)$ while L_2 is the line passing through the points $B\,(4,3,0)$ and $C\,(8,1,-1)$.

(a) Determine parametric equations for L_1 and L_2 (10)

(b) Determine whether L_1 and L_2 are mutually perpendicular [Show all your working] (5)

6.2 Does the point $(8,4,-5)$ lie on the plane $7x - 3y + 4z = 8$? [Show all your working] (3)

6.3 The line L passes through the points $P_1\,(2,4,-1)$ and $P_2\,(5,0,7)$. Determine the point of intersection of L and the xy-plane. (7)

[25]

QUESTION 7

7.1 Determine an equation for the plane that passes through the point (1, 1, 1) and is parallel to the plane

$$x - 3y - 2z - 4 = 0.$$

(3)

7.2 Determine the volume of the parallelepiped in 3-space determined by the vectors $\underline{u} = (1, 3, -1)$, $\underline{v} = (1, 1, 2)$ and $\underline{w} = (3, -1, 2)$. (3)

7.3 Calculate the distance between the plane $2x - 3y + 6z = -1$ and the point $(1, -4, 3)$. (4)

[10]

QUESTION 8

8.1 Express the complex number $1 + i\sqrt{3}$ in polar form. (4)

8.2 (a) State De Moivre's theorem (1)

(b) Use De Moivre's theorem to find all the cube roots of -8 (5)

[10]

TOTAL [100]